

Memorial University of Newfoundland
CMSC 6950: Computer Based Research Tools and Applications
Course Outline – Fall 2025

Instructor: Dr. Scott MacLachlan, Email: smaclachlan@mun.ca . MUN requires that you email me using your @mun.ca email address for anything related to the course. I generally answer course-related emails each weekday, at the beginning and end of the day; if I haven't responded to your email within 2 working days, please resend it. Please do not send me messages on Brightspace, as I do not regularly check these messages.

Meeting times: Tuesdays and Thursdays, 10:30-10:45am, in CSF 2218. This is an on-campus course, with classes held in-person at scheduled times.

Office Hours: Mondays from 1:30-4:00pm and Thursdays from noon-1:30pm, in HH-2019. Other times are available by appointment.

Instructional Continuity: Should courses be disrupted, this class will transition to remote delivery, with readings assigned and discussed via Webex in scheduled (synchronous) sessions, during the class times above. Changes to class schedule will be announced via Brightspace, including cancellations or transitions to online discussions.

Website: All class information will be available on the course's Brightspace shell.

Textbook: Relevant links to content will be posted on the course webpage. There are no recommended/required textbooks to purchase.

Prerequisite: Past experience with some computing and comfort using a computer for scientific tasks is expected. This course will include programming with Python and using the Unix command line interface. Please see me if you have no past computer experience or concerns about your preparation for this course.

Important Dates:

Tuesday, Sept. 23	Last day to add courses
Tuesday, Sept. 30	National Day for Truth and Reconciliation, no classes
Monday-Tuesday, Oct. 13-14	Semester Break, no classes
Tuesday, Nov. 4	Last day to drop courses without academic prejudice
Tuesday, Nov. 11	Remembrance Day, no classes
Wednesday, Nov. 12	Tuesday schedule!
Friday, Dec. 5	Last day of lectures

Copyright: All materials and lectures delivered or provided in CMSC 6950: Computer Based Research Tools and Applications by Scott MacLachlan, including any visual or audio recording thereof, are subject to copyright owned by Scott MacLachlan, unless otherwise indicated. It is prohibited to record, copy, or share by any means, in any format, openly or surreptitiously, in whole or in part, in the absence of express written permission from Scott MacLachlan, any of the lectures or materials provided or published in any form during or from this course.

Health and Safety:

- There is nothing more important than your mental and physical health. You are encouraged to seek appropriate medical attention, as needed, from the Student Wellness and Counselling Centre, and to take advantage of additional academic and student life supports at the university.
- To protect yourself and those around you, it is important to stay home if you feel unwell. Please keep me informed, so that we can work together to allow you to keep up with the course materials should you need to miss classes. You will not be penalized if you need to stay home for any illness.

Evaluation: 50% Assignments, 50% Final Project

Assignments:

- There will be about 7 homework assignments posted to the course Brightspace page, approximately one every 3 classes. Assignments are to be turned by the given deadline. You are allowed two “freebies”, in which you may turn in an assignment 72 hours late and still receive credit; otherwise, late assignments will not be marked. If you require a longer extension due to exceptional circumstances, please provide me with appropriate documentation and a suggested timeline for completing the work.
- For students who are forced to miss classes for acceptable reasons, including having COVID-19 symptoms, due to the contraction of COVID-19, or the requirement to attend to a family member afflicted, this portion of the grade may be determined by averaging over a reduced number of assignments.
- Should there be a significant disruption to the course schedule, the number of assignments may be reduced, but homework will still account for 50% of the final course grade.

Final Project:

- In addition to the homework assignments, there will be a required final project report (of about 10 pages, typeset, including figures and references) and in-class project presentation (of about 5 minutes).
- Details of the final project will be posted to the course Brightspace page in the first week of October. A one-page project proposal will be due mid-October, with feedback provided by the end of October.

Academic Misconduct: Students are expected to adhere to those principles which constitute proper academic conduct. A student has the responsibility to know which actions, as described under Academic Offences in the University Regulations, could be construed as dishonest or improper. Students found guilty of an academic offence may be subject to a number of penalties commensurate with the offence including reprimand, reduction of grade, probation, suspension or expulsion from the University. For more information regarding this policy, students should refer to the Regulations for Academic Misconduct in either Section 6.12 of the University Calendar or Section 4.12 of the School of Graduate Studies Calendar.

The submission of work that has been created by generative artificial intelligence (GAI) tools and presented as a student’s original work is considered an academic offence in this course. Students may not use GAI in any of the following ways:

- Submitting an assignment presented as their own. This includes but is not limited to essays, research papers, computer code, and solutions to mathematical problems.
- Using GAI as an aid to completing an assignment, such as paraphrasing from GAI content.
- Submitting answers to any quiz, test, or exam question using GAI solutions.

Accommodations: Memorial University of Newfoundland is committed to fostering equitable and accessible learning environments for all students. Accommodations for students with disabilities are provided in accordance with the Accommodations for Students with Disabilities Policy (www.mun.ca/policy/site/policy.php?id=239) and its related procedures. Students who feel that they may require formal academic accommodations to address barriers or challenges they are experiencing related to their learning are encouraged to contact Accessibility Services (the Blundon Centre) at the earliest opportunity to ensure any required accommodations are provided in a timely manner. You can contact Accessibility Services (Blundon Centre) by emailing blundon@mun.ca.